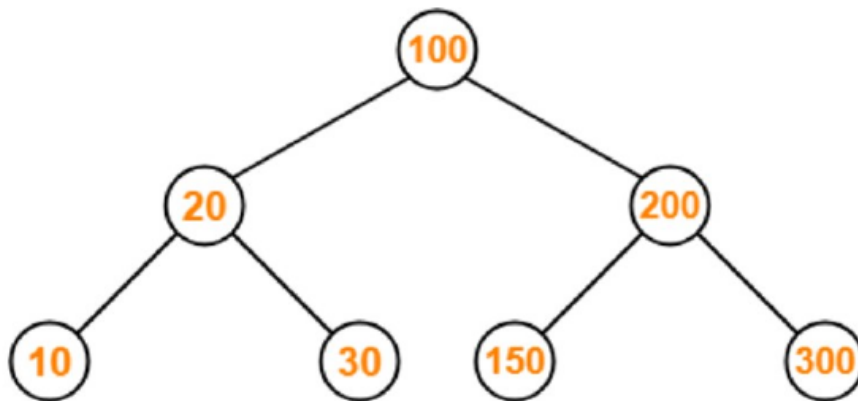


CO3 – AT3 – Tree Traversal Challenges

Question No : 1

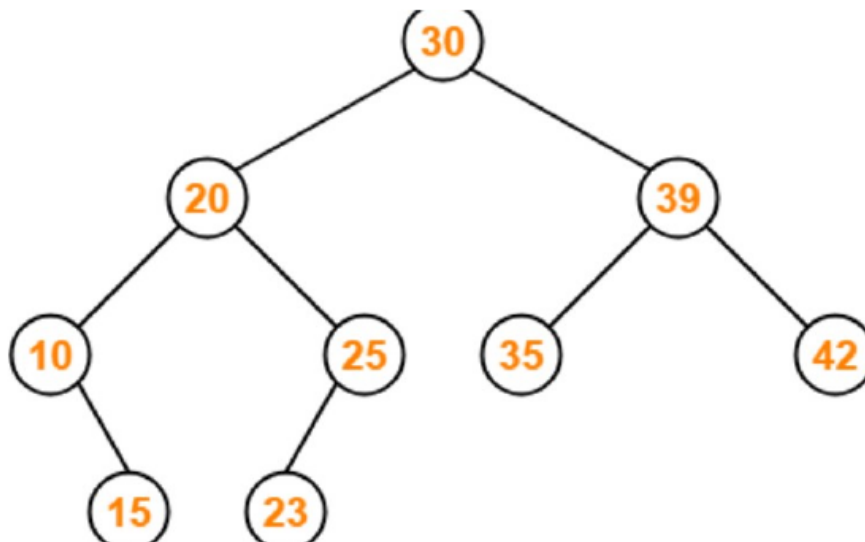
Question No :1



Answer

- Inorder Traversal: 10, 20, 30, 100, 150, 200, 300
- Preorder Traversal: 100, 20, 10, 30, 200, 150, 300
- Postorder Traversal: 10, 30, 20, 150, 300, 200, 100

Question No : 2



Answer

- Inorder Traversal: 10, 15, 20, 23, 25, 30, 35, 39, 42
- Preorder Traversal: 30, 20, 10, 15, 25, 23, 39, 35, 42
- Postorder Traversal: 15, 10, 23, 25, 20, 35, 42, 39, 30

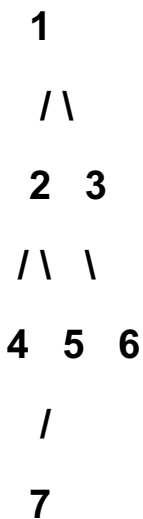
Question No : 3

Assume the digits 7, 5, 1, 8, 3, 6, 0, 9, 4, 2 are added in that sequence into a binary search tree that is initially empty. The binary search tree follows the standard natural number ordering. What is the resulting tree's traversal sequence?

Answer

- The BST formed has 7 as the root; 5 is its left child and 8 its right child. Continuing the insertions gives the required tree.
- Inorder Traversal: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9
- Preorder Traversal: 7, 5, 1, 0, 3, 2, 4, 6, 8, 9
- Postorder Traversal: 0, 2, 4, 3, 1, 6, 5, 9, 8,

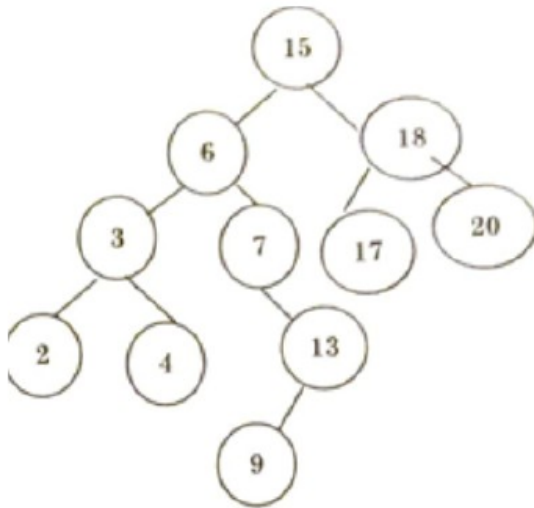
Question No : 5



Answer

- Inorder Traversal: 4, 2, 7, 5, 1, 3, 6
- Preorder Traversal: 1, 2, 4, 5, 7, 3, 6
- Postorder Traversal: 4, 7, 5, 2, 6, 3, 1

Question No : 6



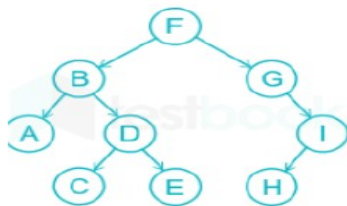
Answer

- Inorder Traversal: 2, 3, 4, 6, 7, 9, 13, 15, 17, 18, 20
- Preorder Traversal: 15, 6, 3, 2, 4, 7, 13, 9, 18, 17, 20
- Postorder Traversal: 2, 4, 3, 9, 13, 7, 6, 17, 20, 18, 15

Answer

Question No 7

What is the sequence of nodes when applying in-order traversal on the binary tree given below?



1. A, B, C, D, E, F, I, H, G
2. A, C, D, E, B, F, G, H, I
3. A, B, C, D, E, F, G, H, I

Answer: A, C, D, E, B, F, H, G, I

- Correct option: 2

Question No 8

Consider the following Inorder and Preorder traversal of a tree

Inorder – D B E F A G H C

Preorder - A B D E F C G H

What is the Postorder Traversal of the same tree

1. D F E B H C G A

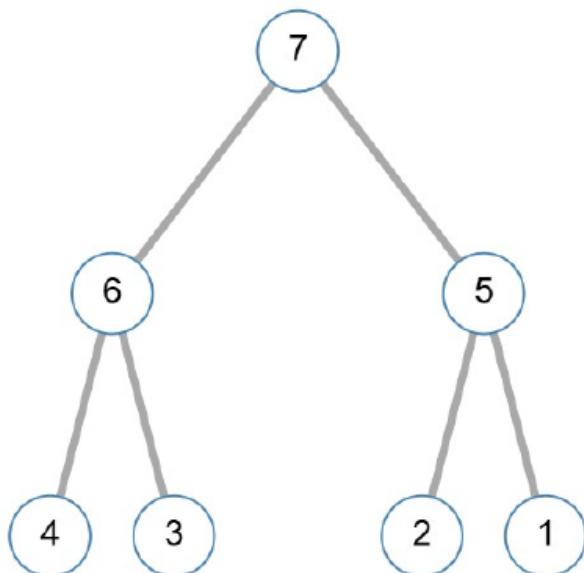
2. D F E B G H C A

3. D F E B H G C A

Answer

- Postorder Traversal: D, F, E, B, H, G, C, A
- Correct option: 3

Question No 9 : Find Inorder, Preorder and Postorder

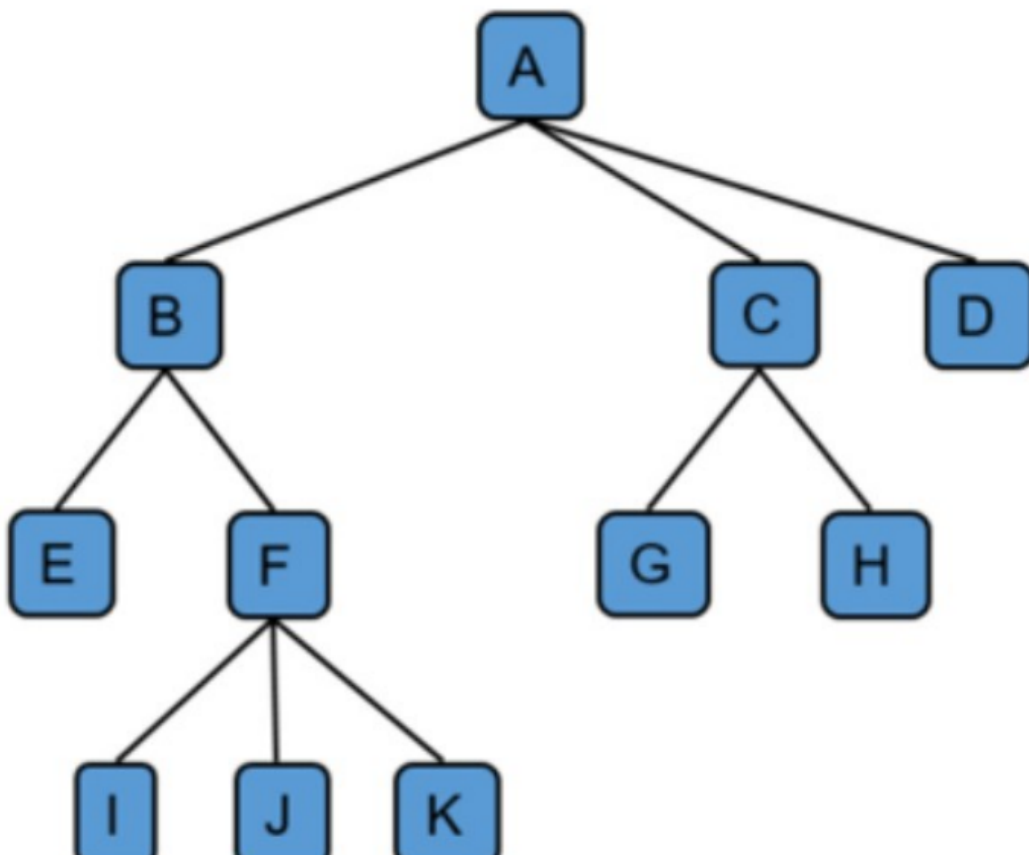


Answer

- Inorder Traversal: 4, 6, 3, 7, 2, 5, 1
- Preorder Traversal: 7, 6, 4, 3, 5, 2, 1
- Postorder Traversal: 4, 3, 6, 2, 1, 5, 7

Question No : 10

Question No 10



Answer

- The uploaded document shows the tree diagram but does not state the traversal type for Question 10.
- Preorder Traversal (left-to-right): A, B, E, F, I, J, K, C, G, H, D
- Postorder Traversal (left-to-right): E, I, J, K, F, B, G, H, C, D, A
- Level-order Traversal: A, B, C, D, E, F, G, H, I, J, K
- Note: Inorder traversal is not uniquely defined for a general tree with more than two children.